

The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

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OFFSITE
2026

EConophysics Lab

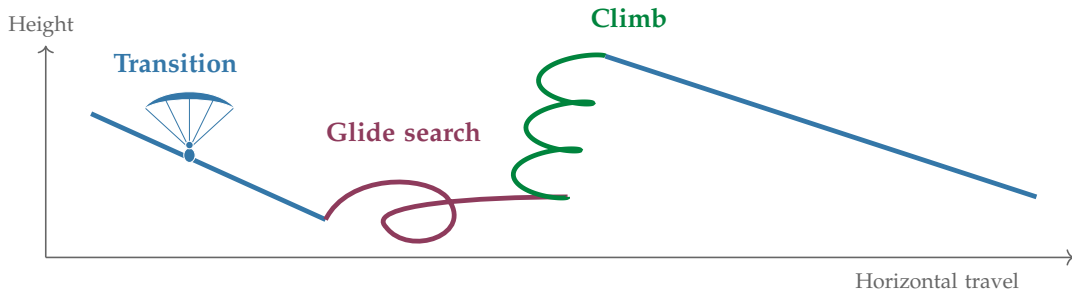
CFM

ILB Institut
Louis
Bachelier

Soaring is transport with an energy budget

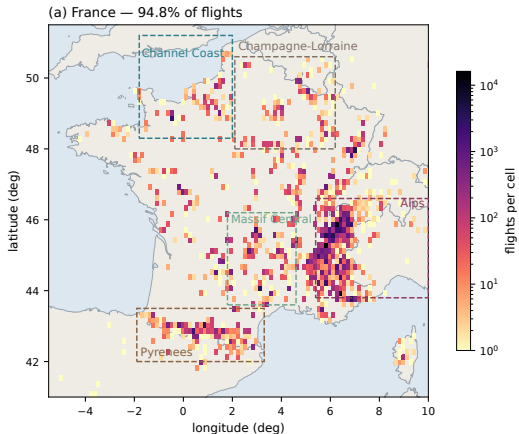
2

Height is spent to travel and recovered in rising air.



Wind, terrain and route choice act throughout the flight.

Which statistical laws emerge from this sequence of behaviours?



FFVL distance competition

186,052

Paraglider tracklogs
1999–2000 to 2025–2026

6,716

Hang-glider tracklogs
2001–2002 to 2025–2026

Regional boxes

- Channel Coast
- Champagne-Lorraine
- Massif Central
- Alps
- Pyrenees

IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

Preprocessing: clean the recorded flight

4

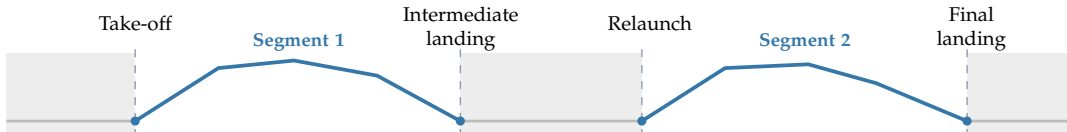
1 GNSS altitude

GNSS height $\longrightarrow$ Trajectory $z(t)$

Barometer $\longrightarrow$ Cleaning checks

GNSS: absolute height without weather drift.
Same high-frequency noise as the barometer.

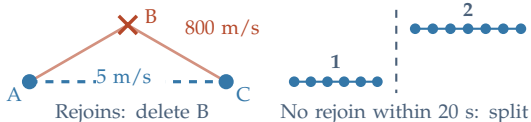
3 Airborne interval



Trim before take-off and after landing.
Stops are not recorded: we infer them from position and altitude.

2 Fix-level cleaning

One fix = (t, φ, λ, z) : time, latitude, longitude, height
Delete an unreachable fix only if the track rejoins without it; otherwise split the flight.



● Retained flight

■ Ground removed

4 Flight selection

Minimum flight thresholds

Time ≥ 40 min

1.13% excluded ($N = 191,108$)

Path ≥ 20 km

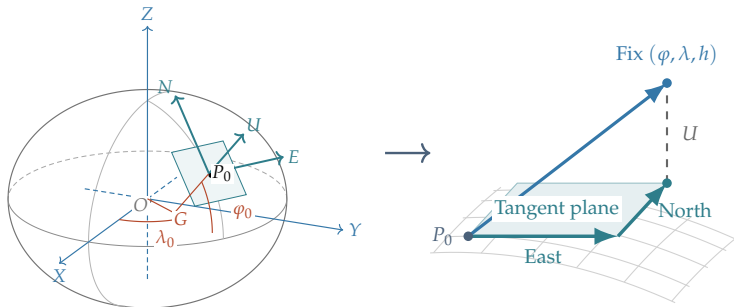
0.09% excluded ($N = 188,942$)

Altitude range ≥ 600 m

2.68% excluded ($N = 188,768$)

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred position
 $\rightarrow$ local east-north distances in metres



Earth model: WGS84 ellipsoid, not a sphere (GPS reference)

P_0 : first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U .

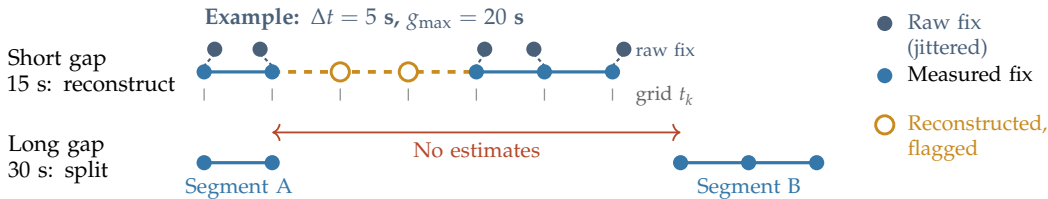
6 Time grid

Put each flight on an exact grid $t_k = t_0 + k\Delta t$ at its own native interval Δt .

- A fix within $\Delta t/2$ of t_k : use it.
- None that close: reconstruct t_k .

Reconstruct only a gap $g \leq g_{\max} = \min(10\Delta t, 20 \text{ s})$.

A longer gap splits the flight into segments.



Keep a segment only if it lasts $\geq 90 \text{ s}$ and $\leq 10\%$ of its grid is reconstructed.

No segment kept: flight dropped. 20,798 paragliders, 193 hang gliders: the largest loss.

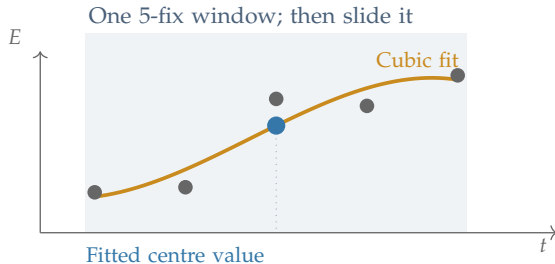
7 Savitzky-Golay fit

Fit a cubic to E , N , z in a moving window, within each segment.

Window: odd, ≥ 5 fixes.

Differentiate the polynomial analytically: no finite differences.

$$\frac{d^n \hat{x}}{dt^n}(t_i) = \left. \frac{d^n P_i}{du^n} \right|_{u=0}, \quad u = t - t_i$$



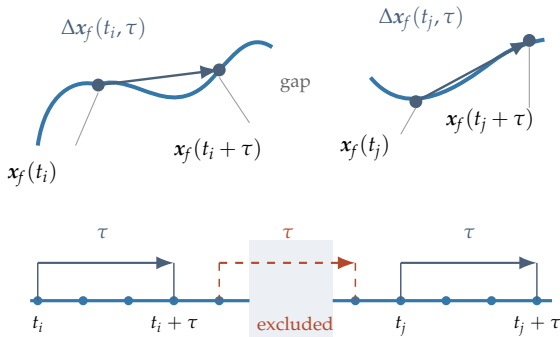
Retained flights after all seven stages (share of downloaded logs)

156,406 paragliders 84.1%

6,094 hang gliders 90.7%

Schematic; the cubic fit is exact for the synthetic fixes shown.

One flight f , two retained segments



- The solid arrows (Δx_f) lie in different segments, but each lies entirely within one: $i, j \in \mathcal{I}_f(\tau)$.
- The dashed arrow crosses the gap and is excluded.

For each lag τ :

- 1 Slide the origin t_i through each segment of f , one grid step at a time, while $t_i + \tau$ stays inside

$$\Delta x_f(t_i, \tau) = x_f(t_i + \tau) - x_f(t_i)$$
- 2 Pool these origins over all segments of f

$$\mathcal{I}_f(\tau) = \{i : t_i, t_i + \tau \text{ in the same segment}\}$$

- 3 Average the $n_f(\tau) = |\mathcal{I}_f(\tau)|$ squared displacements

$$m_f(\tau) = \frac{1}{|\mathcal{I}_f(\tau)|} \sum_{i \in \mathcal{I}_f(\tau)} |\Delta x_f(t_i, \tau)|^2$$

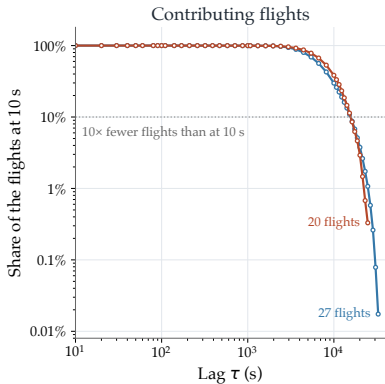
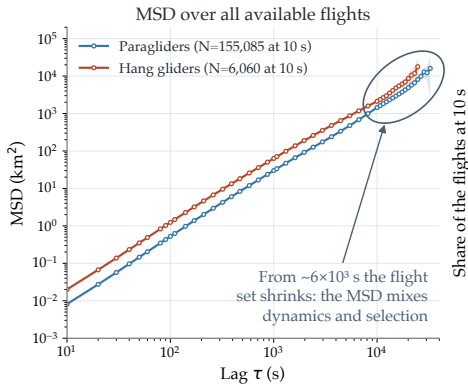
- 4 Average over the $N(\tau)$ flights with at least one origin, each with weight $1/N(\tau)$

$$M_2(\tau) = \frac{1}{N(\tau)} \sum_{f: |\mathcal{I}_f(\tau)| > 0} m_f(\tau)$$

Which flights contribute at each lag?

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All available flights: a flight enters at lag τ if one of its segments has $\tau \leq 0.8 T_s$ ($\geq 1/5$ of it left as origins for m_f).



Desiderata

- 1 The same flights at every lag
- 2 The longest possible lag range
- 3 $N \approx$ of the same order as at 10 s

At long lags only long flights remain, and their number falls steeply.

Any lag-dependent statistic then mixes dynamics with a changing set of flights.

Can we afford a fixed population?

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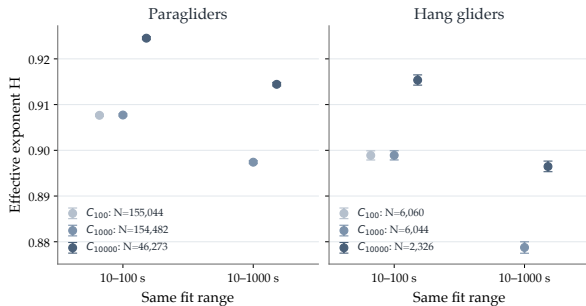
C_{10000} : select once for the largest lag

$\tau_{\max} = 10,000 \text{ s} \leq 0.8 T_s$: $T_s \geq 12,500 \text{ s}$
Only these segments enter, at every lag.

Advantages of C_{10000}

- 1 Same flights and segments, 10 to 10,000 s, with no loss of flights beyond $6 \times 10^3 \text{ s}$
- 2 Three decades of lag; at $\tau_{\max}$ each segment still offers at least 251 origins for m_f
- 3 Sample still large, same order as at 10 s:
46,273 paragliders (30%)
2,326 hang gliders (38%)

Cost of selecting longer flights



Nested cohorts, identical fit ranges, paired bootstrap draws.

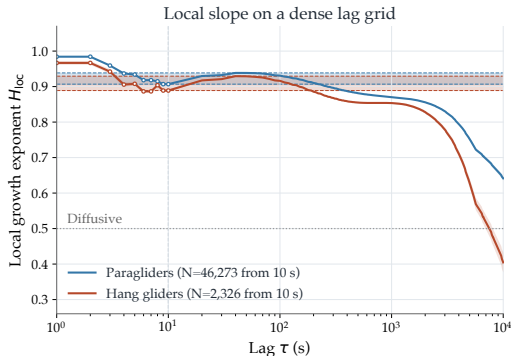
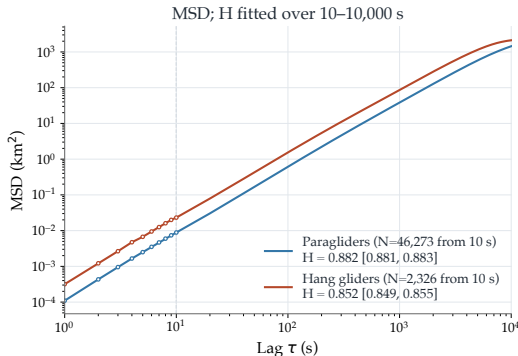
Long flights raise H by 0.016–0.018 on matched lag ranges; origins still decrease with lag.
About 2% in H buys the same flights at every lag over three decades.

Displacement growth from 1 s to 10,000 s

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C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.

$M_2(\tau) \propto \tau^{2H}$: $H = \frac{1}{2}$ diffusion $H = 1$ ballistic motion



$$H_{PG} = 0.882 [0.881, 0.883] > H_{HG} = 0.852 [0.849, 0.855] \text{ over } 10\text{--}10,000 \text{ s}$$

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Which flight characteristics shape MSD and H ?

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Circuit type

Declared route

Open versus **closed**

Terrain / altitude

Proxy: launch altitude

Plains, **hills**, **low** and
high mountains

Experience

Proxy: wing class

Beginners: EN A/B/C
Experts: EN D/CCC

Weather also matters, but we have no data with sufficient spatial and temporal resolution.

Marginal comparison

Example: open vs closed flights
across all altitude classes.

Stratified comparison

Example: open vs closed flights
within each altitude class.

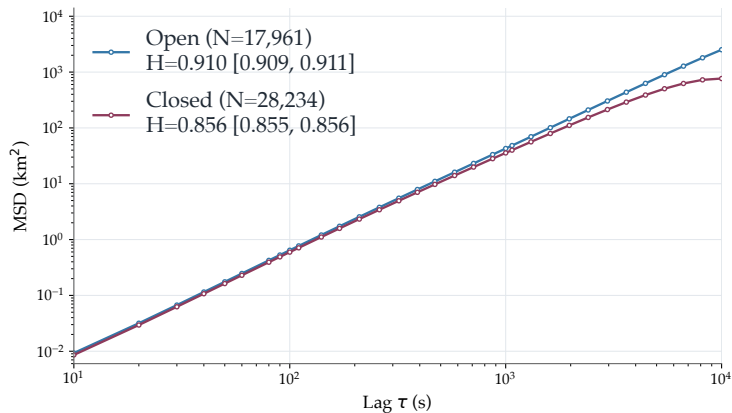
What stratification controls

The altitude class is shared; weather and equipment can still differ between groups.

First compare groups overall, then within the same altitude class.

Open vs closed: how do MSD and H differ?

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Open circuits



17,961 flights

Closed circuits



28,234 flights

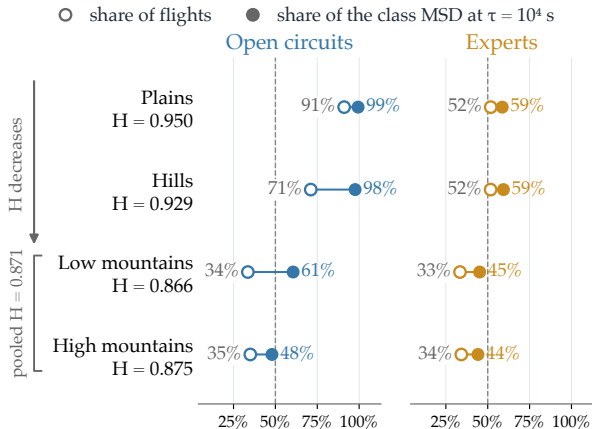
$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] > 0 \quad (90\% \text{ paired interval})$$

Slower closed-route MSD growth is consistent with turns and returns.

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

Does the altitude ranking of H measure terrain alone?

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1 Different mix ○

Lowlands: more open circuits and more Experts, both with higher H .

2 Different weight ●

Group g , a circuit type or wing class, enters as flight share $\times$ MSD:

$$M_{2,a}(\tau) = \sum_g p_{g|a} M_{2,g,a}(\tau)$$

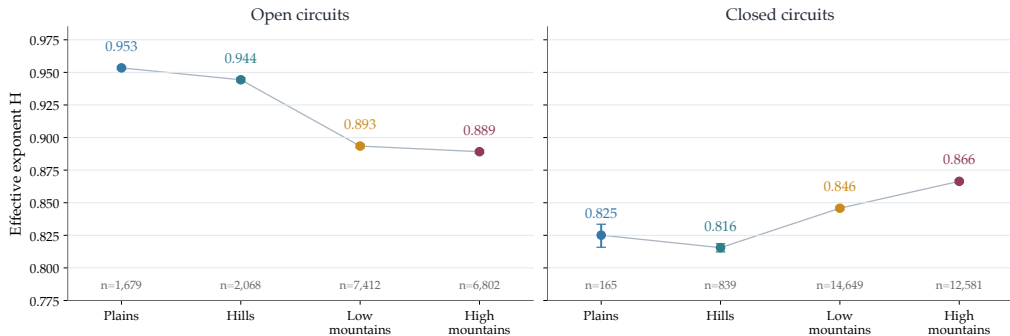
Low mountains: open flights are 34% of flights, 61% of the MSD at 10^4 s. Circuit type shifts the weights more than wing class.

Pooled H : Plains > Hills > mountains; the circuit and wing mix changes with altitude too.
An altitude-only comparison mixes terrain with circuit and wing composition.

Shares of all flights in each class; unclassified: circuit < 0.3%, wing up to 10%. Observed shares, no confidence bands.

Does the open-closed H gap change with altitude?

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Open: lowland H is higher. In mountains, each transition must head for known lift, so the flight moves out less directly.

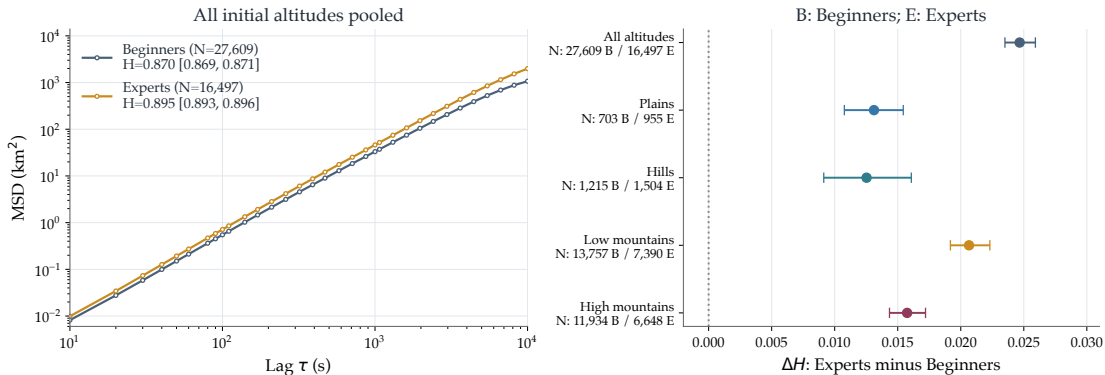
Closed: mountain H is higher. In plains, pilots can turn back sooner and more often; in mountains, even the way home must follow the lift.

Plains: thermals turn up here and there. Mountains: routes must follow ridges and slopes.

Open > closed in every class; mountain orography constrains both, and the gap narrows.

Does the expert–beginner H gap change with altitude?

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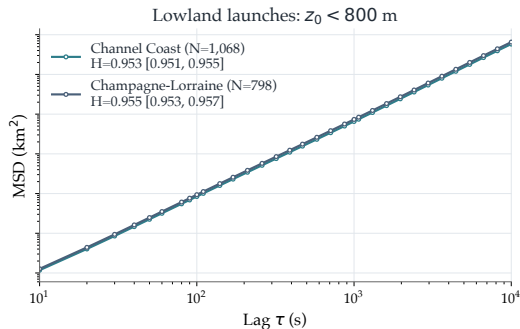
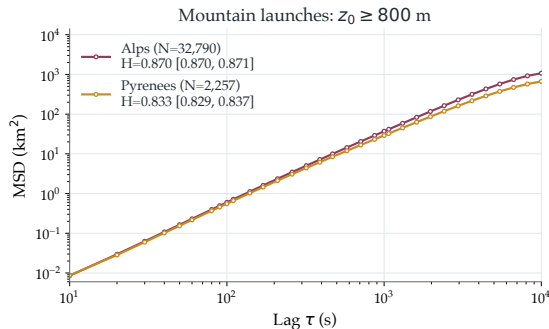
Higher H for experts in every class; the gap depends on altitude.

A constant speed advantage changes MSD amplitude, not H .

Wing class is a proxy. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

Within altitude groups, how does H differ by region?

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Mountain hypothesis

Slope-linked lift ties routes to ridge and valley geometry.

Lowland hypothesis

Roads and dark fields may supply broadly distributed thermal triggers.

ΔH : Alps–Pyrenees 0.037 [0.033, 0.041]; lowlands -0.002 [-0.005 , 0.001].

Similar lowland H ; Pyrenees may constrain routes more than the Alps.

How do climb locations change with height?

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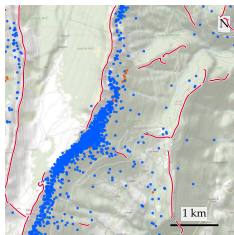
All dates and years pooled. Vilpellet climb crossings.

H2: Chartreuse, Alps

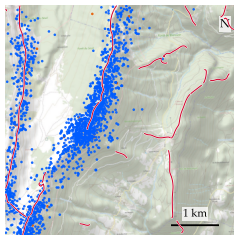
High mountains #2 / 15,840 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,856 crossings / 1,261 flights
PG: 1,249 HG: 12



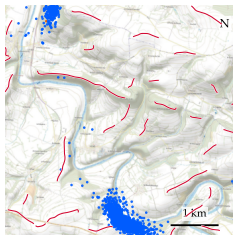
1,868 crossings / 1,452 flights
PG: 1,434 HG: 18

P1: Suisse Normande, Channel Coast

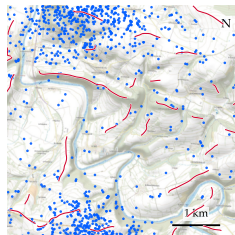
Plains #1 / 2,246 flights

$z_{AGL} = 100 \text{ m}$

$z_{AGL} = 600 \text{ m}$



1,441 crossings / 734 flights
PG: 733 HG: 1



901 crossings / 574 flights
PG: 574 HG: 0

100 to 600 m: P1 climbs spread out; H2 climbs stay near slopes and ridges.

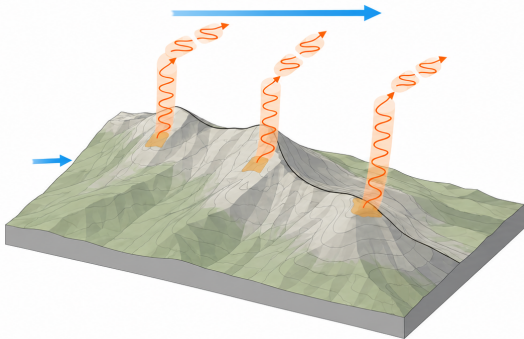
Observed lift use is less confined aloft in P1,
supporting more spatial freedom for route choices.

Why do lowland climb locations spread aloft?

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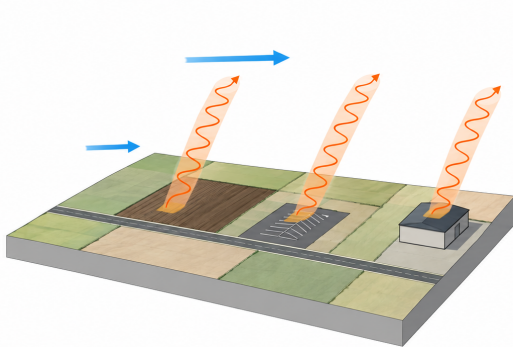
Mountains: sources on heated slopes

Possible shelter below crests; stronger shear aloft.



Lowlands: separated heated surfaces

Dry field, asphalt, warm roof; drift from the base.

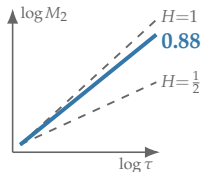


Distributed sources and variable wind could broaden lowland crossings.

Mountain slope sources and possible shelter can keep lift use closer to terrain corridors.

Orange: rising air; blue: wind. Illustrative profiles, not measurements. Sources: FAA, Ch. 9–10; SSA, Lift Sources.

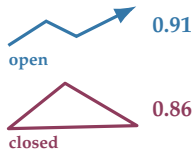
Scaling



Strongly superdiffusive

$H = 0.88$ for paragliders, 0.85 for hang gliders, fitted over $10\text{ s} - 10^4\text{ s}$ on the same flights at every lag.

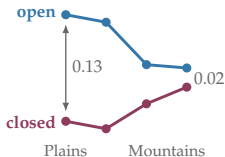
Route and wing



Open routes spread faster

Open exceeds closed in every altitude class (+0.054 pooled). Experts exceed beginners, by less (+0.025).

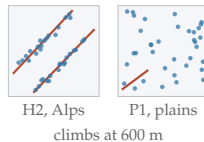
Terrain



Mountains narrow the gap

In mountains, open-route H falls and closed-route H rises. Altitude alone mixes terrain with route and wing mix.

Lift geography



Mountain lift hugs ridges

From 100 to 600 m, H_2 climbs keep to slopes and ridges; P_1 climbs spread out. Observed lift use; causes untested.

Strongly superdiffusive in every group; route and terrain interact, wing adds a small shift

Shifting hypothesis: terrain constrains routes through where pilots can find lift.

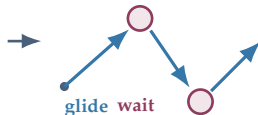
Segmentation



Revise Vilpellet's HMM

Built on ≥ 1 Hz tracks;
misses some of our climbs.
Next: continuous features
in physical time.

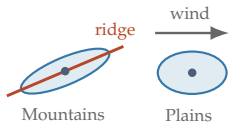
Stochastic model



Describe the observations

Candidate: glides alternate
with waits (search + climb),
built from measured glides,
turns and waiting times.

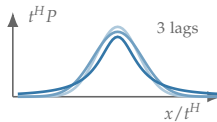
Anisotropy



Ridge axis or wind?

Mountains: chain axis.
Plains: weaker, mean wind.
Preliminary observations.

Distribution scaling



Do distributions collapse?

Collapse and moment tests:
mixed so far. Power-law
MSD alone does not
establish self-similarity.

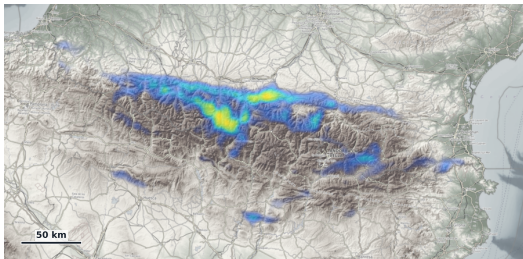
Thank you
for your attention

Where do mountain climbs concentrate?

23

Pyrenees: full chain

Atlantic to Mediterranean



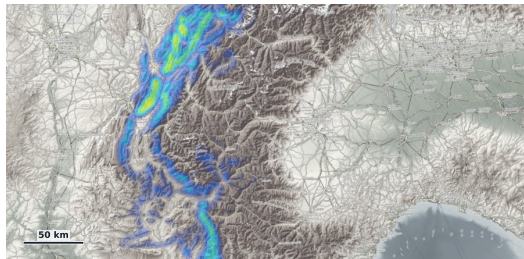
2,257 flights: open 698 (30.9%), closed 1,556 (68.9%)

$$H = 0.833$$

Share of flights with a climb

Western Alps

Matched 450 × 220 km area



32,790 flights: open 10,533 (32.1%), closed 22,212 (67.7%)

$$H = 0.870$$



Pyrenees: a narrow northern band. Alps: more branching climb areas.

Similar open shares make terrain guidance plausible, but the maps do not establish causation.

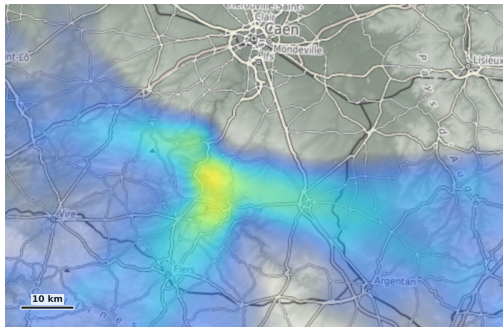
C10000 paragliders; all dates and heights pooled. Equal 450 × 220 km views and colour range. Map: © OSM contributors, SRTM, OpenTopoMap (CC-BY-SA).

How broadly are lowland climbs distributed?

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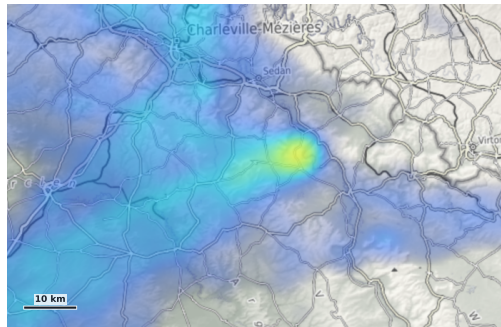
Channel Coast

1,068 flights, $H = 0.953$, 95.5% open



Champagne-Lorraine

798 flights, $H = 0.955$, 92.9% open



Share of flights with a climb

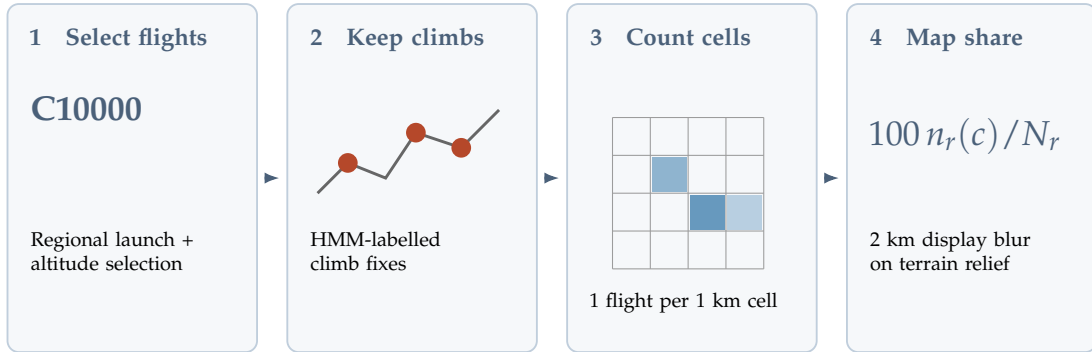


Both windows contain broad patches of climb use, with no single mountain-like crest corridor.

Their similar H values are compatible with weaker spatial constraints in lowlands.

C10000 paragliders; all dates and heights pooled. Matched 100×65 km lowland views; shared colour range. Map: © OSM contributors, SRTM, OpenTopoMap (CC-BY-SA).

One chain for every map: the regional H cohort, with all dates and climb heights pooled.



Colour shows observed climb use, not every available thermal or a causal effect on H .

A: 4 altitudes C: 2 circuits E: beginners / experts $4 \times 2 \times 2 = 16$ cells

$$H_{ace} = \underbrace{\mu + A_a + C_c + E_e}_{\text{additive part}}$$

$$+ (AC)_{ac} + (AE)_{ae} + (CE)_{ce} + (ACE)_{ace}$$

Example: two altitude levels and two circuits; beginners / experts group held fixed.

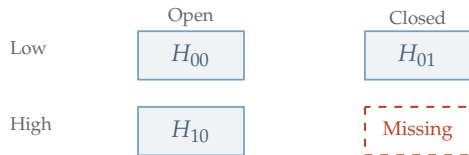
Free coefficients:

6 additive

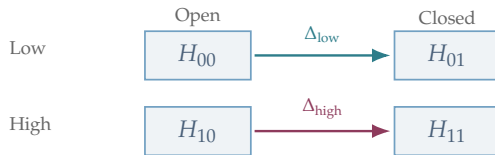
10 interactions

Sum-to-zero constraints

OFAT: one combination missing



Full factorial: all four combinations



Interaction: $D_{AC} = \Delta_{\text{high}} - \Delta_{\text{low}} \neq 0$ means the circuit effect depends on altitude

What a group label requires

Flights in the same small cell, close enough in time to see each other: visual contact, about 15 minutes apart at most.

Then compare transport within shared altitude, circuit and wing-class settings.

$$\Delta H_{a,c,e}^{\text{social}} = H_{a,c,e,\text{group}} - H_{a,c,e,\text{solo}}$$

What the archive gives

Best case: the most populated 5 km cell (Low mountains), busiest day, peak 15 min.

≤ 63 flights

22 July 2023, 13:15–13:30 Paris time.
An upper bound, before any split over 32 cells. Other cells are sparser.

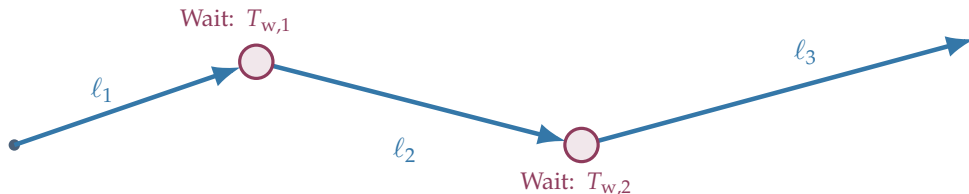
With the current archive, a solo versus group comparison is most likely out of reach

Displacement: gliding

Duration T_g , length ℓ and direction θ .
Retain turning-angle memory between glides.

Waiting: search + thermalling

T_w : total time between successive glides.
Estimate one pooled waiting-time distribution.



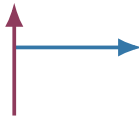
Waiting neglects local search motion and thermal drift.

Monte Carlo: test MSD, displacement distributions, anisotropy and memory

$$\mathbf{v}(t) = s(t) \mathbf{e}(t), \quad \mathbf{e} = (\cos \theta, \sin \theta) \quad \Delta\theta = \theta(t + \tau) - \theta(t)$$



Aligned: $\cos \Delta\theta = 1$



Perpendicular: 0



Opposite: -1

$$C_v(\tau) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle}{\langle s^2 \rangle} = \frac{\langle s(t)s(t + \tau) \cos \Delta\theta \rangle}{\langle s^2 \rangle}$$

$$C_\theta(\tau) = \langle \mathbf{e}(t) \cdot \mathbf{e}(t + \tau) \rangle = \langle \cos \Delta\theta \rangle$$

Constant speed: $C_v = C_\theta$.

Persistent mean drift can leave a nonzero baseline; also inspect centred velocity covariance.

C_v combines speed and direction; C_θ isolates directional agreement

1 Group flights

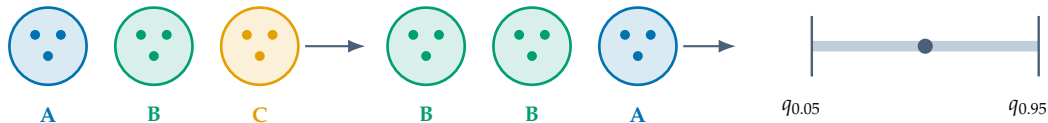
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.

parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

[OpenTopoMap](#): regional relief and contours (OSM / SRTM, CC-BY-SA).

[IGN / Géoplateforme](#): local Plan IGN backgrounds.

[IGN RGE ALTI](#): local terrain and derived crests.

Locator and analysis

[Natural Earth](#): France outline, public domain.

MSD, intervals and crossings: this thesis's prepared pipeline.

[Code, figure scripts and recorded source provenance](#)

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